Lecture 19

Thursday, 23 March 2023 1:36 PM

Wagner No.:

↳ We can make good assumptions about uniformity

$$W_a = \frac{R_{ct}}{R_\Omega}$$

electrodes dealt w/ B-V kinetics

- Charge transfer resistance (R_{ct}) → This is usually at the electrode (since no charge transfer is happening with @ the electrode)
  - ↳ Resistance to current flow due to charge transfer kinetics @ electrodes ($\eta = V - \phi_s - U$)
  - ↳ Last chapter we said $R_{ct} = \left.\frac{\partial(V-\phi)}{\partial i}\right|_{i_{avg}}$ or $\left.\frac{\partial \eta_c}{\partial i}\right|_{i_{avg}}$ ★ ; i_{avg} → average current on the entire electrode

↑ Kinetics → i_{avg} ↑ → R_{ct} ↓ → $W_a \to 0$ (primary)

$$i = i_o \frac{(\alpha_a + \alpha_c) F \eta}{RT} \Rightarrow \frac{\partial \eta}{\partial i}$$

  - ↳ Since dependent on kinetics → will change w/ the given process: · small overpotential (linear kinetics: $R_{ct} = \frac{RT}{(\alpha_a + \alpha_c) F i_o}$)

Anodic Tafel: $\left\{ i = i_o \exp\left[\frac{\alpha_a F \eta}{RT}\right] \right.$

· Large overpotential (Tafel kinetics: $R_{ct} = \left.\frac{\partial \eta}{\partial i}\right|_{i_{avg}} = \frac{RT}{\alpha_a F i_{avg}}$) ↳ does depend on i_{avg}

* Any other type of kinetics → solve for $\eta = f(i)$ & differentiate, or get slope of

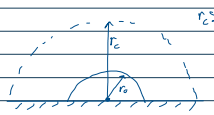
- Ohmic Resistance (R_Ω):
  - ↳ Resistance of current flow due to property of electrolyte + cell

for 1D case, $R_\Omega = \frac{L}{\kappa} = \frac{\text{characteristic length}}{\text{conductivity}}$ ↳ Does not depend on kinetics

↳ Depends on κ (type of electrolyte), L (geometry)

Need to select this carefully

ex:



Hemispherical electrode (spherical coord since it becomes a 1D problem!)

If uniform composition:

$$\nabla^2 \phi = 0$$

$$\Rightarrow \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial \phi}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial \phi}{\partial \theta}\right) + \frac{1}{r^2 \sin^2\theta}\frac{\partial^2 \phi}{\partial \theta^2} = 0$$

$$\Rightarrow \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial \phi}{\partial r}\right) = 0$$

BC's: $\phi = V$ @ $r = r_o$; $\phi = 0$ @ $r = \infty$

$$\therefore \frac{d^2\phi}{dr^2} + \frac{2}{r}\frac{d\phi}{dr} = \frac{r}{2}\frac{\partial^2 \phi}{\partial r^2} + \frac{\partial \phi}{\partial r} = 0$$

$$\phi = \frac{c_1}{r} + c_2 \xrightarrow{BC's} \boxed{\phi = \frac{r_o V}{r}}$$

$$i = -\kappa \left.\frac{d\phi}{dr}\right|_{r=r_o} \quad (\nabla c = 0)$$

$$i_\Omega = \frac{V}{r_o}\kappa \rightarrow i_\Omega = \text{ohmically limited current (controlled by potential & not kinetics)}$$

$$R_\Omega = \frac{V}{i_\Omega} \text{ (ignoring c.t.)}$$

$$\therefore \boxed{R_\Omega = \frac{r_o}{\kappa}}$$ ∴ This is the characteristic length.

ex: for hemicylindrical electrode

$$R_\Omega = \frac{r_o}{\kappa} \ln\left(\frac{r_{CE}}{r_o}\right)$$ r_{CE} = counter electrode

↓

Need this for W_a

↓

Need W_a for current dist.

↓

change values for r_{CE} and r_o to determine R_Ω dependence (what dominates)

Putting together R_{ct} and R_Ω: $W_a = R_{ct,L} = \frac{RT}{F}\kappa$) Linear

Putting together R_{ct} and R_Ω: $W_{a,L} = \frac{R_{ct,L}}{R_\Omega} = \frac{RT}{(\alpha_a+\alpha_c)F i_o l}$ } Linear kinetics

$W_{a,T} = \frac{R_{ct,T}}{R_\Omega} =$